

Ch. 7 Estimation

7.1 Statistical Inference

- I. Probability & Statistical Models
 - A. Statistical Model
 - 1. Identification of r.v.'s (both observable & hypothetical)
 - 2. Specification of a joint dist'n (or family of joint dist'ns) for observable r.v.'s
 - 3. Identification of unknown parameter(s)
 - 4. Specification of joint dist'n for unknown parameter(s)
 - θ r.v. $\Rightarrow$ joint dist'n of observable r.v.'s indexed by θ is $f(x_1, \dots, x_n | \theta)$.
 - B. Statistical Inference
 - 1. Probabilistic statement procedure that produces a probabilistic statement about some or all parts of a statistical model.
 - Ex.* Produce r.v. $Y = g(X_1, \dots, X_n)$ s.t. $P(Y \geq \theta | \theta) = 0.9$.
 - C. Parameter / Parameter Space
 - 1. Parameter of dist'n: characteristic(s) that determine the joint dist'n for the r.v.'s of interest
 - 2. Parameter space: set Ω of all possible values of $(\theta_1, \dots, \theta_k)$
- II. General Classes of Inference Problems
 - A. Prediction
 - 1. Predicting r.v.'s that have not yet been observed
 - 2. When unobserved quantity to be predicted is a parameter, prediction is called estimation
 - B. Statistical Decision Problems
 - 1. After experimental data has been analyzed, choose a decision from available class of decisions, with property that consequences of each available decision depend on the unknown value of some parameter
 - Ex.* Hypothesis testing.
 - C. Experimental Design
 - 1. Close relationship between design of an experiment; statistical analysis of experimental data
- III. Definition of a Statistic
 - A. Suppose $X_1, \dots, X_n$ are observable r.v.'s. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}$. Then $T = r(X_1, \dots, X_n)$ is a statistic.

7.2 Prior & Posterior Dist'ns

- I. Prior Dist'n
 - A. Suppose we have statistical model w/ parameter θ . θ is r.v. $\Rightarrow$ distribution before observing other r.v. is called prior dist'n, $\xi(\theta)$.
- II. Posterior Dist'n
 - A. $\xi(\theta | x) = \frac{\xi(\theta) f(x | \theta)}{g(x)}$, $x \in \mathbb{R}^n$, $\theta \in \Omega$.
 - B. $g(x) = \begin{cases} \sum_{\theta \in \Omega} \xi(\theta) f(x | \theta), & \Omega \text{ countable,} \\ \int_{\Omega} \xi(\theta) f(x | \theta) d\theta, & \text{else.} \end{cases}$
- III. Likelihood Function
 - A. Since $g(x)$ is not function of θ , $\xi(\theta | x) \propto \xi(\theta) f(x | \theta)$.

- B. When joint density $f(x | \theta)$ of observations in a random sample is a function of θ , define $f(x | \theta) := L(\theta | x)$.
- C. “Posterior dist’n of θ is proportional to the product of the likelihood function & prior dist’n of θ .”
- IV. Sequential Observations & Prediction
 - A. A random sample $X_1, \dots, X_n$ can only be observed sequentially:

$$\xi(\theta | x) \propto f(x_n | \theta) \xi(\theta | x_1, \dots, x_{n-1}).$$

7.3 Conjugate Prior Dist’n

I. Sampling from Bernoulli Dist’n

- A. $\theta \sim \text{BETA}(\alpha, \beta)$, $X_1, \dots, X_n$ iid $\text{BER}(\theta)$

$$\Rightarrow \theta | x \sim \text{BETA}\left(\alpha + \sum_{i=1}^n x_i, \beta + n - \sum_{i=1}^n x_i\right).$$

- B. (α, β) are prior hyperparameters; $(\alpha + \sum x_i, \beta + n - \sum x_i)$ are posterior hyperparameters.

II. Sampling from Poisson Dist’n

- A. $\theta \sim \text{GAM}(\alpha, \beta)$, $X_1, \dots, X_n$ iid $\text{POI}(\theta)$

$$\Rightarrow \theta | x \sim \text{GAM}\left(\alpha + \sum_{i=1}^n x_i, \beta + n\right).$$

III. Sampling from Normal Dist’n

- A. $\theta \sim N(\mu_0, v_0^2)$, $\bar{X} | \theta \sim N\left(\theta, \frac{\sigma^2}{n}\right)$, σ^2 known

$$\Rightarrow \theta | x \sim N\left(\frac{\sigma^2 \mu_0 + n v_0^2 \bar{x}}{\sigma^2 + n v_0^2}, \frac{\sigma^2 v_0^2}{\sigma^2 + n v_0^2}\right).$$

IV. Sampling from Exponential Dist’n

- A. $\theta \sim \text{GAM}(\alpha, \beta)$, $X_1, \dots, X_n$ iid $\text{EXP}(\theta)$

$$\Rightarrow \theta | x \sim \text{GAM}\left(\alpha + n, \beta + \sum_{i=1}^n x_i\right).$$

V. Improper Prior Dist’n

- A. Let $\xi : \Omega \rightarrow \mathbb{R}^+$ s.t. $\int_{\Omega} \xi(\theta) d\theta = \infty$. If $\xi(\theta)$ comes from prior dist’n of θ , then ξ is called an improper prior for θ .

7.4 Bayes Estimators

I. Nature of Estimation Problem

- A. Estimation: let $X_1, \dots, X_n$ be observable data whose joint density is indexed by parameter $\theta \in \Omega \subset \mathbb{R}^d$.
- B. Estimator: $\hat{\theta} = \delta(X)$ is an r.v.
- C. Loss function: $L : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^+$.
- Goal: $\hat{\theta} := \arg \min_{\delta} \{E_{\theta}[L(\theta, \delta)]\}$, $E_{\theta}[L(\theta, \delta)] = \int_{\Omega} L(\theta, \delta(x)) \xi(\theta) d\theta$.

II. Definition of Bayes Estimator

A. If x observed,

$$\hat{\theta}_{\text{Bayes}} = \arg \min_{\delta} \{E_{\theta}[L(\theta, \delta) | x]\}, \quad E_{\theta}[L(\theta, \delta) | x] = \int_{\Omega} L(\theta, \delta(x)) \xi(\theta | x) d\theta.$$

III. Different Loss Functions

A. Squared loss

1. $L(\theta, \delta) = (\theta - \delta)^2$
2. $E(\theta | x) < \infty \Rightarrow \hat{\theta}_{\text{Bayes}} = E(\theta | x)$ “posterior mean”

B. Absolute loss

1. $L(\theta, \delta) = |\theta - \delta|$
2. $\hat{\theta}_{\text{Bayes}} = \eta(\theta | x)$ “posterior median”

IV. Bayes Estimate for Large Samples

A. Effect of different prior dist'ns

1. When n large, Bayes estimators not too different; priors are not same

B. Consistency

1. $\hat{\theta}$ is consistent $\Rightarrow \hat{\theta} \rightarrow \theta$
2. If (i) $X_1, \dots, X_n$ form random sample from $f(x | \theta)$, (ii) conjugate prior assigned to θ , (iii) $L(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2$, then $\hat{\theta}_{\text{Bayes}}$ is consistent.

V. More General Parameters

A. $h : \Omega \rightarrow \mathbb{R}^d \Rightarrow \hat{h}(\theta) = \arg \min_{\hat{h}} \{E_{\theta}[L(h(\theta), \hat{h}(\theta)) | x]\}$

B. $L(h(\theta), \hat{h}(\theta)) = (h(\theta) - \hat{h}(\theta))^2 \Rightarrow \hat{h}_{\text{Bayes}}(\theta) = E_{\theta}[h(\theta) | x]$

VI. Limitations of Bayes Estimator

A. Specification of prior & loss function may be incorrect

B. $\dim(\Omega) > 1 \Rightarrow$ difficult to assign multivariate prior dist'n

7.5 Maximum Likelihood Estimators

I. Definition

- A. $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \{f(x | \theta)\}$
- B. $\hat{\theta}_{\text{MLE}}(x) \in \Omega$
- C. $\max_{\theta} f(x | \theta)$ DNE $\Rightarrow \hat{\theta}_{\text{MLE}}$ DNE; $\max_{\theta} f(x | \theta)$ not unique $\Rightarrow \hat{\theta}_{\text{MLE}}$ not unique

II. Examples

- A. X iid BER(θ) $\Rightarrow \hat{\theta}_{\text{MLE}} = \bar{X}_n$
- B. X iid GEO(θ) $\Rightarrow \hat{\theta}_{\text{MLE}} = (\bar{X}_n + 1)^{-1}$ ($x_i = 0, 1, 2, \dots$)
- C. X iid POI(θ) $\Rightarrow \hat{\theta}_{\text{MLE}} = \bar{X}_n$
- D. X iid UNI($0, \theta$) $\Rightarrow \hat{\theta}_{\text{MLE}} = \max\{X_1, \dots, X_n\}$
- E. X iid EXP(θ) $\Rightarrow \hat{\theta}_{\text{MLE}} = (\bar{X}_n)^{-1}$
- F. X iid $N(\mu, \sigma^2) \Rightarrow (\hat{\mu}_{\text{MLE}}, \hat{\sigma}_{\text{MLE}}^2) = (\bar{X}_n, \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2)$

III. Limitations of MLE's

A. MLE might not exist

B. MLE might not be unique

C. $\hat{\theta}_{\text{MLE}}$ doesn't represent “most likely” value of θ given the data

7.6 Properties of MLE's

I. Invariance

A. $g : \Omega \rightarrow \Omega'$ and $\hat{\theta}_{\text{MLE}}$ exists

- B. $g(\hat{\theta}_{\text{MLE}})$ exists if $g(\hat{\theta}_{\text{MLE}}) \in \Omega'$
- II. Consistency
 - A. Under general assumptions (smooth likelihood func. & unique $\hat{\theta}_{\text{MLE}}$), $\hat{\theta}_{\text{MLE}} \rightarrow \theta$
- III. Numerical Computation
 - A. Newton's Method
 - B. Gradient Descent
- IV. Method of Moments
 - A. Can be used as initial guess for Newton's method
 - B. Definition
 1. $X_1, \dots, X_n$ iid $f(x | \theta)$, $\theta \in \Omega \subset \mathbb{R}^k$
 2. Let $\mu_j = E(X^j)$, $j = 1, \dots, k$
 3. $\mu : \Omega \rightarrow \mathbb{R}^k$, $\mu = (\mu_1, \dots, \mu_k)$
 4. $M : \mathbb{R}^k \rightarrow \Omega$, $M(\mu(\theta)) = \theta$
 5. $\hat{\mu}_j = \frac{1}{n} \sum_{i=1}^n X_i^j$, $j = 1, \dots, k$
 $\Rightarrow \hat{\theta}_{\text{MM}} = M(\hat{\mu}) = M(\hat{\mu}_1, \dots, \hat{\mu}_k)$.
 - C. $\hat{\theta}_{\text{MM}}$ is consistent, but sometimes biased

7.? MLE–Bayes Estimator Relationship

- A. Let γ_n be a function of the prior hyperparameters s.t. $\gamma_n \rightarrow 1$ as $n \rightarrow \infty$.
- B. $E(\theta | x) = \gamma_n \hat{\theta}_{\text{MLE}} + (1 - \gamma_n)E(\theta)$.
- C. If $L(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2$, then

$$\hat{\theta}_{\text{Bayes}} = \gamma_n \hat{\theta}_{\text{MLE}} + (1 - \gamma_n)\mu_\theta.$$

7.7 Sufficient Statistics

- I. Definition
 - A. Suppose $X = (X_1, \dots, X_n)$ is a random sample from dist'n f indexed by $\theta \in \Omega \subset \mathbb{R}^k$.
Let $T = r(X)$ be a statistic.
 - B. T is sufficient statistic for θ iff for all values of the statistic, the conditional dist'n of the data given the statistic and θ is the same for all θ .
 - C. T s.s. for $\theta \iff f_{X|T}(x | t, \theta) = h(x, t)$, $\forall t, \theta$.
- II. Factorization Criterion
 - A. N–F Factorization Thm: T s.s. for $\theta \iff f_X(x | \theta) = u(x)v(r(x), \theta)$, $u, v \geq 0$.
 - B. Bayesian Definition: T s.s. for $\theta \iff \xi(\theta | x) \propto v(r(x), \theta)\xi(\theta)$.

7.8 Jointly Sufficient Statistics

- I. Definition
 - A. Suppose $\theta \in \Omega \subset \mathbb{R}^k$, $T \in \mathbb{A} \subset \mathbb{R}^k$. $(T_1, \dots, T_k)$ jointly sufficient statistics for θ iff $f_{X|T}(x | t, \theta) = h(x, t)$, $\forall t, \theta$.
 - B. N–F Thm (Extended): let $r : S^n \rightarrow \mathbb{R}^k$ be vector-valued statistic. $T = r(X)$ is j.s.s. for $\theta \iff f_X(x | \theta) = u(x)v(r(x), \theta)$ for all θ .
- II. Minimal s.s.
 - A. Order statistics $(X_{(1)}, \dots, X_{(n)})$, $Y_1 = \min\{X_1, \dots, X_n\}$, $Y_n = \max\{X_1, \dots, X_n\}$
 - B. $Y_1, \dots, Y_n$ s.s. for θ
 - C. Minimal (jointly) s.s. for θ : if $T_j^* = g_j(T)$, $\forall j = 1, \dots, k$, T' is some arbitrary s.s.

- D. S is m.s.s. for θ iff $\frac{f(x | \theta)}{f(y | \theta)} = h(x | y) \iff S(x) = S(y)$
E. $\hat{\theta}_{\text{MLE}}$ s.s. for $\theta \Rightarrow \hat{\theta}_{\text{MLE}}$ m.s.s. for θ
F. $\hat{\theta}_{\text{Bayes}}$ s.s. for $\theta \Rightarrow \hat{\theta}_{\text{Bayes}}$ m.s.s. for θ

7.9 Improving an Estimator

- I. Mean squared error of Estimator
A. $R(\theta, \hat{\theta}) = E_{\theta}[(\hat{\theta} - h(\theta))^2]$, $\hat{\theta} = \hat{\theta}(X)$.
II. Conditional Expectation when a Sufficient Statistic is Known
A. $\hat{\theta}^* = E(\hat{\theta} | T)$, T is s.s. for $\theta \in \Omega$.

$$\Rightarrow R(\theta, \hat{\theta}^*) \leq R(\theta, \hat{\theta}), \quad \forall \theta \in \Omega.$$

B. If $R(\theta, \hat{\theta}) < \infty$ and $\hat{\theta} \neq g(T)$, then $R(\theta, \hat{\theta}^*) < R(\theta, \hat{\theta})$.
C. Inadmissible: $\hat{\theta}$ is inadmissible if $\exists \hat{\theta}_0$ s.t. $R(\theta, \hat{\theta}_0) \leq R(\theta, \hat{\theta})$, $\forall \theta \in \Omega$, and $\exists \theta_0$ s.t. $R(\theta_0, \hat{\theta}_0) < R(\theta_0, \hat{\theta})$.
D. $\hat{\theta}_0$ dominates $\hat{\theta}$.
E. Admissible: $\hat{\theta}$ is admissible if no $\hat{\theta}_0$ dominates $\hat{\theta}$.
F. Restatement: “ $\hat{\theta} \neq g(T) \Rightarrow \hat{\theta}$ is inadmissible” $\Leftrightarrow$ “ $\hat{\theta}$ admissible $\Rightarrow \hat{\theta}$ is s.s. for $\theta \in \Omega$.”
III. Limitation of Use of S.S.
A. S.S. highly dependent on choice of density function modeled

Ch. 8 Sampling Dist'n of Estimators

8.1 Sampling Dist'n of a Statistic

- I. Statistics & Estimators
A. Sampling dist'n: $X = (X_1, \dots, X_n)$ iid $f(x | \theta)$, $\theta \in \Omega$, $T = r(X)$

$$\Rightarrow f_T(t | \theta) \text{ is sampling dist'n.}$$

8.2 Chi-Square Dist'n

- I. Def'n
A. Let $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$, $\sigma^2 \in \mathbb{R}^+$. Let $\hat{\sigma}_0^2 = \hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$, μ known.
B. $X \sim \chi_m^2 \iff X \sim \text{GAM}(m/2, 1/2)$

$$f(x) = \frac{x^{m/2-1} e^{-x/2}}{2^{m/2} \Gamma(m/2)}, \quad x > 0, \quad m \in \mathbb{N}.$$

- II. Properties
A. $M_X(t) = (1 - 2t)^{-m/2}$, $t < 1/2$
1. $E(X) = m$
2. $V(X) = 2m$
B. $X_1, \dots, X_n$ indep., $X_i \sim \chi_m^2 \Rightarrow \sum_{i=1}^n X_i \sim \chi_{nm}^2$
C. $X_1, \dots, X_n$ iid $N(0, 1) \Rightarrow \sum_{i=1}^n X_i^2 \sim \chi_n^2$
D. $n\hat{\sigma}_0^2/\sigma^2 \sim \chi_n^2 \Rightarrow n\hat{\sigma}_0^2/\sigma^2 \sim \text{GAM}(n/2, 1/2)$
E. $\hat{\sigma}_0^2 \sim \text{GAM}(n/2, n/(2\sigma^2))$

8.3 Joint Dist'n of Sample Mean and Sample Variance

I. Suppose $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$, $(\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$. Let $S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$. Then

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{(n-1)S_n^2}{\sigma^2} \sim \chi_{n-1}^2.$$

$$\hat{\mu} = \bar{X}_n, \quad \hat{\sigma}^2 = S_n^2/n, \quad \hat{\sigma}^2 \sim \text{GAM}\left(\frac{n-1}{2}, \frac{n}{2\sigma^2}\right).$$

II. Orthogonal Matrices Supplement (used to prove statement I)

A. **Def1:** A is orthogonal $\iff A \in O_n = \{A \in \mathbb{R}^{n \times n} \mid A^T A = I_n\}$

B. **Def2:** A is orthogonal $\iff A \in \{A \in \mathbb{R}^{n \times n} \mid A = (u_1, \dots, u_n), u_i \cdot u_j = 0, i \neq j, u_k \in \mathbb{R}^n\}$

C. Properties:

1. $|\det(A)| = 1, \forall A \in O_n$

2. Preserves distances: $\|AX\| = \|X\|, \forall A \in O_n$

3. $X_1, \dots, X_n$ iid $N(0, 1)$, $Y = AX, \forall A \in O_n \Rightarrow Y_1, \dots, Y_n$ iid $N(0, 1)$

8.4 The t -Dist'n

I. Definition

A. $X \sim t_m \iff X = \frac{Z}{\sqrt{Y/m}}, Y \sim \chi_m^2, Z \sim N(0, 1)$

B. pdf:

$$f_X(x) = \frac{\Gamma\left(\frac{m+1}{2}\right)}{\sqrt{m\pi} \Gamma\left(\frac{m}{2}\right)} \left(1 + \frac{x^2}{m}\right)^{-(m+1)/2}, \quad x \in \mathbb{R}.$$

C. moments: $E(|X|^k)$ DNE for $m \leq 1$ or $m \leq k$

D. $m \in \mathbb{N} \Rightarrow$ first $m-1$ moments exist; $E(|X|^m)$ DNE

II. Relation to Cauchy Dist'n

A. $X \sim t_1 \iff X \sim \text{Cauchy}, f_X(x) = \frac{1}{\pi(1+x^2)}, x \in \mathbb{R}$

B. $E(X^k)$ DNE, $\forall k \in \mathbb{N}$

III. Relation to Normal Dist'n

A. $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$, $r = \sqrt{S^2/(n-1)}$

$$\Rightarrow \sqrt{n} \frac{\bar{X}_n - \mu}{r} \sim t_{n-1}.$$

IV. Relation to Standard Normal Dist'n

A. $X_n \sim t_n \Rightarrow X_n \xrightarrow{d} N(0, 1)$

B. Uses fact $\lim_{n \rightarrow \infty} \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right) \sqrt{n/2}} = 1$

8.5 Confidence Intervals

I. C.I. provide method of adding more information to an estimate $\hat{\theta}$ by creating random interval (A, B) that has high probability containing unknown θ .

II. C.I. for Mean of Normal Dist'n

A. Def:

1. $X_1, \dots, X_n$ iid $f(x \mid \theta), \theta \in \Omega$

2. $g : \Omega \rightarrow \mathbb{R}$
 3. A, B statistics s.t. $P(A < g(\theta) < B) = \gamma$
 $\Rightarrow (A, B)$ is $100\gamma\%$ C.I. for $g(\theta)$.
 4. $P(A < g(\theta) < B) = \gamma$ called exact C.I.
 5. (a, b) is observed value of C.I. after $X_1, \dots, X_n$ has been observed
- B. Normal dist'n w/ unknown σ^2

$$X_1, \dots, X_n \text{ iid } N(\mu, \sigma^2) \Rightarrow \forall \gamma \in (0, 1) :$$

$$A = \bar{X}_n - t_{n-1}^{-1} \left(\frac{1+\gamma}{2} \right) \frac{s}{\sqrt{n}}, \quad B = \bar{X}_n + t_{n-1}^{-1} \left(\frac{1+\gamma}{2} \right) \frac{s}{\sqrt{n}}.$$

C. Interpretation of C.I.

1. (A, B) : before observing data, we can be $100\gamma\%$ confident that (A, B) contains $g(\theta)$
2. After repeated sampling from same population we expect $100\gamma\%$ of these C.I.'s to contain the true parameter $g(\theta)$

III. One-Sided C.I.

A. Def:

1. Lower $100\gamma\%$ C.I. for $g(\theta)$: (A, ∞) s.t. $P(A < g(\theta)) \geq \gamma$
 2. Upper $100\gamma\%$ C.I. for $g(\theta)$: $(-\infty, B)$ s.t. $P(g(\theta) < B) \geq \gamma$
- B. $P(A < g(\theta)) = \gamma$ or $P(g(\theta) < B) = \gamma$ called exact lower/upper C.I.
- C. Normal dist'n w/ unknown σ^2

$$X_1, \dots, X_n \text{ iid } N(\mu, \sigma^2) \Rightarrow \forall \gamma \in (0, 1) :$$

$$A = \bar{X}_n - t_{n-1}^{-1}(\gamma) \frac{s}{\sqrt{n}}, \quad B = \bar{X}_n + t_{n-1}^{-1}(\gamma) \frac{s}{\sqrt{n}}.$$

IV. C.I. for Other Parameters

A. Pivotal

1. $X = (X_1, \dots, X_n)$ iid $f(x | \theta)$, $\theta \in \Omega$
 2. $V = V(X, \theta)$ is pivotal if:
 - a) f_V same $\forall \theta$
 - b) $\exists g$ s.t. $V(V(X, \theta), X) = g(\theta)$
- B. If V is a pivotal w/ cont. cdf F_V and $r(v, x)$ strictly increasing in v for all x , then the following are $100\gamma\%$ C.I. endpoints:

$$A = r^{-1}(\gamma_1, x), \quad B = r^{-1}(\gamma_2, x), \quad \gamma_2 - \gamma_1 = \gamma.$$